

AGHAR USMAN KANNANTHODI

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PROFESSIONAL SUMMARY

Backend developer with internship experience building production REST APIs, LLM-powered pipelines, and ETL/data-engineering systems. Strong foundation in Python, SQL, and distributed data workflows, with hands-on experience designing batch ETL pipelines, schema migrations, and query-performance optimization. Currently building Data Engineering depth in PySpark, Databricks, and Snowflake, and holds the Databricks Fundamentals Badge. Looking to grow into a Data Engineering role building scalable pipelines and data platforms.

TECHNICAL SKILLS

Data Engineering (Learning): PySpark, Databricks, Snowflake, ETL Pipeline Design, Batch Processing, Data Modeling

Languages: Python (Advanced), SQL, JavaScript, Java, C

Databases: PostgreSQL, MSSQL, MySQL, SQLite, pyodbc, psycopg2

Backend & APIs: Flask, FastAPI, REST APIs, Microservices, Rate Limiting

Cloud & DevOps: GCP, Docker, Linux/Unix, Git/GitHub, CI/CD

AI/ML & NLP: Ollama, Llama 3, Mistral, BERT, Transformers, PyTorch, TensorFlow, Scikit-learn, HuggingFace

Data & Tools: Pandas, NumPy, NLTK, spaCy, TF-IDF, FuzzyWuzzy, OpenCV, React.js

CERTIFICATIONS

- **Databricks Fundamentals Badge** — Databricks Training

PROFESSIONAL EXPERIENCE

Python Backend Intern

Feb 2025 – May 2025

Team Thai — Calicut, Kerala

- Designed and built a Text-to-SQL chatbot using Llama 3 and Mistral (via Ollama) with a dual-model fallback architecture, 3-attempt retry logic, and thread-safe TTL caching to handle concurrent requests reliably.
- Engineered SQL injection prevention with 7-pattern regex detection and SELECT-only enforcement, securing natural language queries over an MSSQL sales database.
- Built a Flask REST API with a thread-pool executor for concurrent request handling, configurable pagination (10–500 rows), and CSV export; optimized query response time by ~35% through caching and batch fetching.
- Designed and implemented a PostgreSQL ETL pipeline for a 22-column customer schema — batch insertion (100 rows/batch), per-row retry logic, timestamp normalization, and automated failed-row reporting.

TECHNICAL PROJECTS

AgriLeaf Pro — CNN-based Plant Disease Detection with IoT

2024 – 2025

Integration

GitHub

- Designed and trained a 5-layer CNN classifying 19 plant diseases across 4 crops, achieving 85%+ accuracy with sub-100ms inference on test hardware.
- Built a Flask REST API with OpenCV image preprocessing, secure token-based authentication, and automated treatment recommendation responses.
- Integrated a ThingSpeak IoT data pipeline ingesting 6 real-time soil parameters with 7-day historical visualization.
- Stack: TensorFlow, Flask, OpenCV, ThingSpeak, GCP

ACHIEVEMENTS & ADDITIONAL INFORMATION

- **State-Level Hackathon — 2nd Place:** Led a 4-member team in a 6-hour hackathon; owned backend architecture, coordinated task split, and delivered a working product under time pressure.
- **Languages:** English (Professional), Hindi (Professional), Malayalam (Conversational), Kannada (Conversational)